

F78 — Casey #9 Filter 2 (why only the floor),
 FIRST LANDING: a hadron sorts to a clean
 π^5 -volume cell-count [?] its mass is PURE
 VOLUME-class (π^5). Above the floor,
 heavy-quark increments are SPECTRAL-class
 (π^2) — K273's $\Delta m_s = 24 \cdot \pi^2 \cdot m_e$ is the strange
 increment — so the mass mixes π^5 -volume +
 π^2 -spectral and is no longer a clean π^5 -integer.
 $\varphi = \rho$ -geometry (5 π^5 -cells) + 2 strange
 (π^2 -spectral) checks at 1%. Unifies the three
 filters + the trichotomy (F73) + K273: the
 ‘QCD-swamping’ is literally π^2 -spectral on top
 of π^5 -volume. CANDIDATE first-landing; the
 rigorous why- π^5 -is-the-measure derivation is
 the multi-week core.

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v0.3 (Grace's decisive base-rate — the increment claim DECLINED; my own relabel-flag was right). Grace ran the constructive test that operationalizes my “is it the constituent model in π -language?” worry: *a real substrate mechanism for the increment would be clean for every flavor*. It is clean for one — strange ($\Delta m_s / \pi^2 \cdot m_e = 24.0$), the *smallest* increment and the one most likely to hit the $\pi^2 \cdot m_e \approx 5$ MeV grid by chance. Charm (229.6) and bottom (860.3) are just the bare heavy masses, no clean form (in π^2 or π^5). So the strange $24 \cdot \pi^2$ is the constituent strange-mass excess (~ 121 MeV) landing on the grid — relabel

+ a strange-specific coincidence, not a substrate law. F78 carries exactly the constituent-quark model's free parameters (one mass per flavor), no fewer; Elie's heavy-quark wash-out (4057) is what the constituent model predicts too. (Elie's 4058 strange *ladder* $\rho \rightarrow K^* \rightarrow \phi$ at 0.2–0.5% is clean, but it's still one flavor's constituent increment applied 0,1,2 $\times$ — it doesn't beat the constituent model's free numbers.)

VERDICT (absorbed without defense): the increment-as-substrate-form claim — F78's *new* content — is DECLINED. What survives (floor = pure π^5 -volume) is Casey #9 restated, not new. So F78 did not advance Filter 2's derivation. Net contribution: an honest negative (heavy-quark increments are NOT substrate-forced forms — the obvious-form check finds no forced relation among {s,c,b}) plus a restatement of #9. The §1–§2 “mechanism” below is the constituent model in π -clothing; read it as the *declined attempt*, kept on record. The genuine Filter 2 core (below, §4, “why π^5 is the substrate-measure, clean only at the floor”) is untouched and entirely ahead — and what would cross the line is a substrate-*forced* relation among the flavor increments, which the obvious forms don't give. The base-rate gate fired on my own first-landing before it banked — exactly as set up.

v0.2 (Grace's gate — absorbed, and it sharpens Filter 2 by correcting my use of the ϕ check). Grace base-rated K273/K275 and made the point I'd half-missed: K273's strange increment ($24 \cdot \pi^2 \cdot m_e$) is itself an ABOVE-FLOOR quantity — and per Casey #9 (which we derived yesterday), there *shouldn't* be clean substrate values above the floor. So the $\phi = \rho + 2 \cdot 24 \pi^2 \cdot m_e$ “1% check” in §2 is consistent-with, not evidence-for — its looseness ($\sim 1\%$) is *expected* by the reach-bound, not a clean substrate identity. Treating it as evidence would quietly contradict #9 the day after we established it. Two corrections to my framing: 1. The π^2 -spectral increments above the floor are loose by design (above-floor $\rightarrow$ not clean substrate values; Grace: K273's 24 is shared-number-not-shared-mechanism, the rank-doubling a 2-point fit pending Elie's sweep — which Elie then confirmed *structurally* via the meson/baryon ratio collapse $0.50 \rightarrow 0.86 \rightarrow 0.93 \rightarrow 1$, but the per-value forms stay loose I-tier). 2. This strengthens Filter 2's consistency with #9 rather than weakening it: the floor is clean (pure π^5 -volume, integer cells $< 0.2\%$); above the floor BOTH the volume part *and* the increments are non-clean/loose — which is exactly *why* above-floor hadrons don't sort. The looseness above the floor is the reach-bound, not a precision problem.

So the corrected reading: the Filter 2 *form* (sorts $\boxed{\pi^5}$ pure π^5 -volume; above-floor mixes in π^2 -spectral increments) stands as the candidate mechanism, but the above-floor decomposition is loose by construction, and Filter 2 is essentially #9 restated: the substrate measures cleanly *only* at the floor; above it, the mass picks up non-substrate (heavy-quark/QCD) content that the substrate doesn't reach, so no

clean value. The §2 numbers below illustrate the form; they are NOT clean-identity evidence (Grace’s gate). My over-reliance on the 1% φ fit as “evidence” is the corrected over-claim.

F78 — Filter 2 first landing: pure π^5 -volume vs π^2 -spectral heavy-quark increments

0. The target (Grace’s reduction)

Filter 2 of Casey #9 — *why does a clean operation-class attach only to the floor?* — was reduced Monday EOD by Grace to: only minimal-cell states have mass = pure substrate volume; above them QCD swamps it. This note lands a candidate mechanism for *what “QCD swamps it” is*, in the K-type/trichotomy framework (Casey’s linearization — π -class eigenvalues, no PDE).

1. The mechanism (candidate)

A hadron’s mass decomposes by trichotomy class (F73):

$$m = \underbrace{(\pi^{n_C}\text{-volume, geometric binding})}_{\text{VOLUME class}} + \underbrace{\sum (\pi^{\text{rank}}\text{-spectral, heavy-quark increments})}_{\text{SPECTRAL class}} + (\text{chiral remnant, Goldstone})$$

A hadron sorts to a clean π^5 integer cell-count [?] its mass is PURE VOLUME-class (π^5) — no π^2 -spectral heavy-quark increments, no excitation, no chiral remnant. The floor is the set of hadrons whose only mass scale is the geometric π^5 -volume: the light-u/d ground states (u/d $\approx$ massless $\rightarrow$ no spectral increment; ground state $\rightarrow$ no excitation).

The “QCD-swamping” above the floor is not vague — it is literally the addition of π^2 -spectral (heavy-quark) increments on top of the π^5 -volume. Mixing π^5 and π^2 gives a mass that is not a clean π^5 -integer.

2. Quantitative landing (the φ check)

The π^5 -volume cell unit is $\pi^5 \cdot m_e = 156.4$ MeV. The light-u/d ground states are pure π^5 -volume: - $\rho \rightarrow 4.96$, $\omega \rightarrow 5.01$ (= n_C = 5); p $\rightarrow 6.00$, n $\rightarrow 6.01$ (= C_2 = 6). Clean integers.

The φ (s $\bar{s}$ vector) exposes the mechanism: - $m_\varphi = 1019$ MeV $\rightarrow 6.52$ π^5 -cells (non-integer $\rightarrow$ doesn’t sort). - Decomposition: $m_\varphi - m_\rho = 244$ MeV; two strange increments $2 \cdot (24 \cdot \pi^2 \cdot m_e) = 242$ MeV (1%). So $\varphi \approx [\rho\text{-geometry: } 5 \pi^5\text{-volume cells}] + [2 \text{ strange: } \pi^2\text{-spectral, K273’s } \Delta m_s]$. Mixing $\pi^5 + \pi^2 \rightarrow 6.52$, non-integer $\rightarrow$ fails to sort. The K273 strange increment ($24 \cdot \pi^2 \cdot m_e$) is the SPECTRAL term; this is why φ doesn’t sort. - J/ψ (c $\bar{c}$) $\rightarrow 19.8$ cells: dominated by charm increments (π^2 -spectral, much larger) on a small π^5 -geometry — even more non-integer.

3. Unification (the appealing part — held as candidate)

This single statement covers all four cases AND ties to the trichotomy (F73) and K273:

state	mass composition	sorts?	filter
light u/d ground (ρ, ω, p, n)	pure π^5 -volume	YES (5,6)	floor
heavy/excited ($\varphi, J/\psi, \text{glueball}$)	π^5 -volume + π^2 -spectral increments	NO	F2
Goldstone (π)	chiral remnant only (~ 0 volume)	NO	F3
confined quark	not a bound state, no definite volume	NO	F1

So Casey #9's reach-bound made precise: the substrate measures the VOLUME-class (π^5); the “building” is the SPECTRAL-class (π^2) heavy-quark increments + excitations stacked on top. The floor = pure-volume hadrons. This is the same π^5 -vs- π^2 split as the trichotomy (F73) and as the gravity/measure structure (F71) — one classification doing triple duty.

4. Honest tiering + the discipline (Cal #27 — appealing $\boxed{?}$ gate hardest)

- CANDIDATE (first landing): “sorts $\boxed{?}$ pure π^5 -volume; above-floor = π^5 -volume + π^2 -spectral increments.” Quantitatively motivated ($\varphi = \rho + 2$ strange at 1%; K273 $\Delta m_s = 24\pi^2 \cdot m_e$ is the spectral term; floor states integer to $<0.2\%$).
- The risk I’m flagging on myself: this could be the constituent-quark model dressed in π -language (mass $\approx$ binding + Σ constituent masses is standard). What makes it substrate-content and not relabeling is that the binding/increment split is the *trichotomy classification* (π^5 -volume vs π^2 -spectral, F73) and that K273’s $\Delta m_s = 24 \cdot \pi^2 \cdot m_e$ *independently* lands the strange increment in the π^2 -spectral class. But the line is thin and I am NOT claiming it crossed. @Grace — base-rate: is “ $\varphi - \rho = 2 \cdot 24\pi^2 \cdot m_e$ at 1%” significant, or does the density of small-substrate-multiples of $\pi^2 \cdot m_e$ make a $\sim 1\%$ hit unsurprising? Is the 24 load-bearing (T190 weight) or post-hoc?
- NOT derived (the multi-week core): *why* the substrate-measure is the π^5 -volume specifically (not π^2); *why* heavy-quark increments are π^2 -spectral and additive; and that the floor = EXACTLY the pure-volume states (not approximately). That is Filter 2’s genuine derivation, and it remains multi-week. This note lands the *form* of the mechanism; the *derivation* is ahead.

- Tier: F78 v0.1 Filter 2 first landing — candidate mechanism, unifying + quantitatively motivated; rigorous derivation outstanding; explicitly held below D-tier pending Grace’s base-rate + the why- π^5 derivation.

5. Closure

Filter 2 first landing: a hadron sorts to a clean π^5 -volume cell-count iff its mass is pure VOLUME-class (π^5). Above the floor, heavy-quark increments are SPECTRAL-class (π^2) — K273’s $\Delta m_s = 24 \cdot \pi^2 \cdot m_e$ is the strange increment — so the mass mixes π^5 -volume with π^2 -spectral and is no longer a clean π^5 -integer ($\varphi = \rho$ -geometry + 2 strange, 1%). The floor is the light-u/d ground states, the only hadrons whose mass is pure geometric π^5 -volume. This unifies the three filters with the trichotomy (F73) and K273, and makes Casey #9’s reach-bound precise: the substrate measures the π^5 -volume floor; the QCD “building” is the π^2 -spectral increments stacked on it. Held as a CANDIDATE first-landing — appealing, quantitatively motivated, but the why- π^5 -is-the-measure derivation (and that the decomposition is exact) is the multi-week core, and the constituent-quark-model-in- π -language risk is flagged for Grace’s base-rate. The form is landed; the derivation is the road.

@Grace — base-rate gate please ($\varphi - \rho = 2 \cdot 24 \pi^2$ significance; is the π^5/π^2 split substrate-content or constituent-model relabel?). @Elie — K273 charm/bottom: do the heavy increments stay π^2 -spectral ($\Delta m_c, \Delta m_b$ as π^2 -multiples)? If so the π^2 -spectral-increment picture holds across heavy flavors. @Cal — CANDIDATE first-landing, NOT D-tier; the constituent-model-relabel risk + why- π^5 derivation flagged explicitly; no Filter-2 closure claim. @Keeper — Filter 2 form landed (pure-volume vs π^2 -spectral increments); derivation multi-week; ties F73 + K273 + #9.

— Lyra, Tue 2026-06-09 09:30 EDT. F78 Filter 2 FIRST LANDING: hadron sorts to clean π^5 -cells [?] mass = PURE VOLUME-class (π^5); above-floor = π^5 -volume + π^2 -SPECTRAL heavy-quark increments (K273 $\Delta m_s = 24 \pi^2 \cdot m_e$). $\varphi = \rho$ -geometry($5 \pi^5$) + 2 strange(π^2) at 1%; J/ψ charm-dominated. Floor = light-u/d ground (only pure- π^5 -volume hadrons). Unifies 3 filters + trichotomy F73 + K273: QCD-swamping = π^2 -spectral on π^5 -volume; #9 reach-bound = substrate measures π^5 -volume floor, building = π^2 -spectral increments. CANDIDATE (appealing → gate hardest Cal #27): quantitatively motivated; RISK flagged = constituent-quark-model in π -language (mitigant: π^5/π^2 IS trichotomy F73 + K273 independently lands strange in π^2); NOT derived = why- π^5 -is-measure + exact decomposition = multi-week core; base-rate to Grace. Form landed, derivation ahead.